MILITARY SPECIFICATION SHEET

ELECTRON TUBE, THYRATRON

TYPES 7665 AND 7665A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Hydrogen, ceramic-metal

See figures 1 and 1A

Mounting position: any

Weight: 10 ounces nominal

ABSOLUTE-RATINGS:

Parameter:	Ef	epy	epx	Ebb	egy	egx	ib	Ecc	Ip	Ib
Unit:	Vac	kv	kv	Vdc	v	v	a	Vdc	Aac	Adc
Maximum:	6.8	16.0 (Note 1)	16.0 (Note 2)	---	600 (Note 3)	200	350	150	6.5	0.50
Minimum:	5.8	2.0	5% epy	1,000	200	---	---	---	---	---
TEST CONDITIONS:	6.3	16.0	---	---	150	---	---	0	---	---

ABSOLUTE-RATINGS:

Parameter:	prr	Eres	Pb	tk	dik/dt	TA	tj	Cooling
Unit:	pps	Vac	---	sec	a/μs	°C	μs (Note 7)	(Note 5)
Maximum:	Note 4	6.8	5.0×10^9 (Note 4)	---	2,000	+150	0.005	---
Minimum:	---	5.8	---	180	---	-55	---	---
TEST CONDITIONS:	1,000	6.3 (Note 6)	---	180	---	---	---	---

GENERAL:

Qualification - Required

(D) denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
3267	Instantaneous starting	10	epv = 16 kv (min); Ef = Eres = 6.8 V	0.65	II	---	---	---	---
3246	Operation (1)	9	epy = 18 kv; Ef = Eres = 5.8 V; t = 10 minutes	0.65	II	egy	---	150	v
3247	DC anode voltage for conduction	-	Ef = Eres = 5.8 V	0.65	II	Ebb	---	1,000	Vdc
3241	Heater current (cathode)	-		0.65	II	If	3.5	8.0	Aac
3241	Heater current (reservoir)	-		0.65	II	Ires	1.0	4.0	Aac
3251	Pulse emission (method A)	-	ik = 350 a; tp = 5.0 μs ±10%; prr = 60 ±10%; tr = 0.5 μs (max); starting time of pulse = 2.5 μs; specified time interval 4.0 μs	0.65	II	egk	---	200	v
	Quality conformance inspection, part 2								
3246	Operation (1A)	-	Operation (1); t = 10 minutes; tk = 5 minutes; Ef = Eres = 6.8 V	---	---	egy	---	150	V
3256	Anode delay time	-	Operation (1); t = 120 seconds	---	---	tad	---	0.4	μs
3256	Anode delay time drift	11	Anode delay time	---	---	Δtad	---	0.10	μs
3261	Time jitter	9, 12	Operation (1), except epy = 8 kv	---	---	tj	---	0.005	μs
3246	Operation (2)	13	prr = 300; epy = 6.6 kv; t = 20 minutes; Ef = Eres = 5.8 V	---	---	egy	---	150	v
	Quality conformance inspection, part 3								
---	Life test (1)	9	Group C; t = 96 hours "on" and 1 hour "off" (tube mounted horizontally); t = 500 hours	---	---	---	---	---	---
---	Life-test (1) end points:								
3246	Operation (1) and (1A)	-		---	---	egy	---	150	v
3247	DC anode voltage for conduction	-	egy = 150 v	---	---	Ebb	---	1,000	Vdc
3261	Time jitter	-	egy = 150 v	---	---	tj	---	0.005	μs

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
1041	Shock	15	100 G at 11 ms; no voltage applied	---	---	---	---	---	---
1031	Variable-frequency vibration	8, 15	No voltage applied	---	---	---	---	---	---
---	Shock and variable-frequency vibration end points:								
3246	Operation (1)	15		---	---	egy	---	160	v
3261	Time jitter	15		---	---	tj	---	0.005	μs
3246	Operation at elevated ambient temperature	9, 14	TA = 150°C; t = 5 hours	---	---	egy	---	150	v
---	Life test (2)	15	epy = 6.0 kv; prr = 300; t = 500 hours	---	---	---	---	---	---
---	Life-test (2) end points:								
3246	Operation (2)	-		---	---	egy	---	160	v
3247	DC anode voltage for conduction	-	egy = 160 v	---	---	Ehb	---	1,000	Vdc

NOTES:

1. Instantaneous starting is permissible. The maximum permissible epy is 16 kv and shall not be attained in less than 0.04 second.

2. In pulsed operation, the peak inverse voltage, exclusive of a spike of 0.05 μs (maximum) duration, shall not exceed 5.0 kv during the first 25 μs following the pulse.

3. The driver pulse, measured at the tube socket with thyratron grid disconnected: 200 volts minimum, 600 volts maximum; tr = 0.35 maximum; grid pulse duration 2.0 μs minimum. Impedance of drive circuit 50 to 500 ohms. At -55 C, a minimum of 250 volts is required.

4. The tube is capable of operation at more than 50,000 pps within the limitations of the Pb factor and the Ib current ratings. With a saturable reactor, Pb equal to 7×10^9 is permissible for certain application.

5. It may be desirable to employ forced-air cooling under conditions of high Pb number operations. A cooling air-blast of 5 cfm may be directed into the anode cup.

6. The optimum reservoir voltage for operation in accordance with operation (1) conditions is 6.3 Vac and shall be held within ±7-1/2 percent. Applications involving other operating conditions may necessitate the redetermination of the optimum reservoir voltage.

7. Appreciably less jitter than 0.005 μs can be realized using an anode voltage of 8 kv or more, a grid drive amplitude near the maximum and a grid drive impedance near the minimum values.

8. There shall be no pronounced resonance in the range from 0 to 2,000 pps.

9. The circuit constants shall be so chosen, that at epy = 16 kv under resonant charging conditions, dik/dt 1,500 a/μs (min), ib 175 a (min), tp = 1.0 ±10 percent μs, prr 1,000 (min). The grid pulse characteristics shall be: tp = 2.0 μs (max); tr = 0.35 μs (min); and Zs = 500 ohms (min).

10. The tube shall operate satisfactorily on push-button starting within three attempts when the anode voltage (epy) is applied to the tube under test in such a manner as to rise from 0 to 16 kv minimum within 0.03 second. (The filter in the rectifier shall be so designed that the epy reaches at least 7 kv within 0.015 second.) Any tube failing to start within three attempts will be considered a failure.

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NOTES: -Continued

11. This test shall be performed simultaneously with the operation (1) test. An anode delay time measurement shall be made at the end of 2 and 10 minutes of that test. The change in anode delay time (with respect to the 2 minute reading) shall not exceed the specified value.

12. The tube shall be tested by applying a peak forward anode voltage not to exceed the value specified in the test conditions for the time jitter test immediately after the cathode warmup period (tk). After 60 seconds of operation the variation in firing time (tj), measured at 50 percent of the cathode current pulse, shall be not greater than the amount specified.

13. The circuit constants shall be so chosen that under resonant charging conditions at epy 6.0 kv; ib 12.0 a minimum; dik/dt 150 a/μs minimum; tp = 66 μs ± 10 percent; prr = 400; inverse voltage due to mismatch - 10 percent epy. The grid pulse characteristics shall be: tp 2.0 μs (max); tr 0.35 μs (min); internal driver impedance 500 ohms (min).

14. This test shall be conducted for a total of 5 hours with no more than three kickouts. The tube shall be started with 107.5 percent Eres and operate at this value for 4 hours. At the start of the fifth hour, and while the tube is still operating, the voltage shall be lowered to Eres = 92-1 2 percent and remain there for the final 1 hour of operation.

15. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of two. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 0.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Air Force - 11, 80
DIA - ES

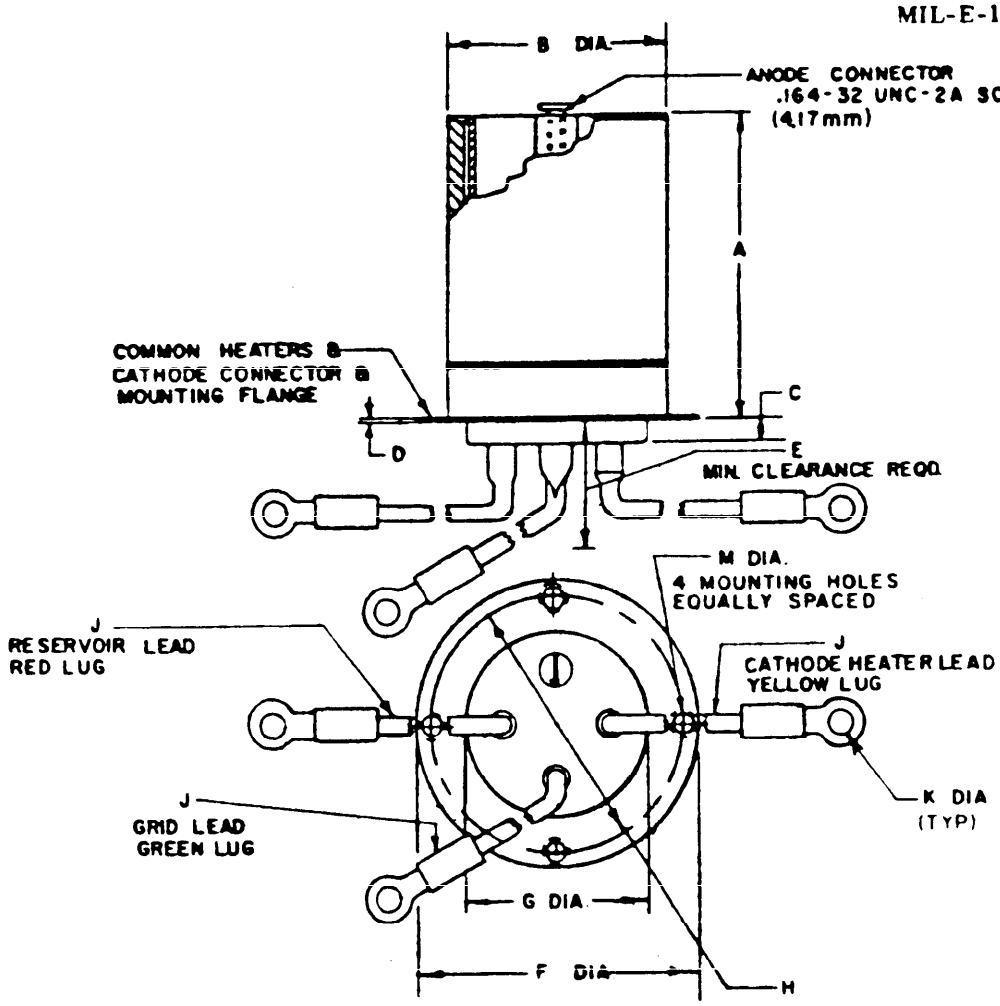
User activities:
Army - AR
Navy - AS, OS, CG, SH

Preparing activity: Navy - EC

Agent: DLA - ES

(Project 5960-3217)

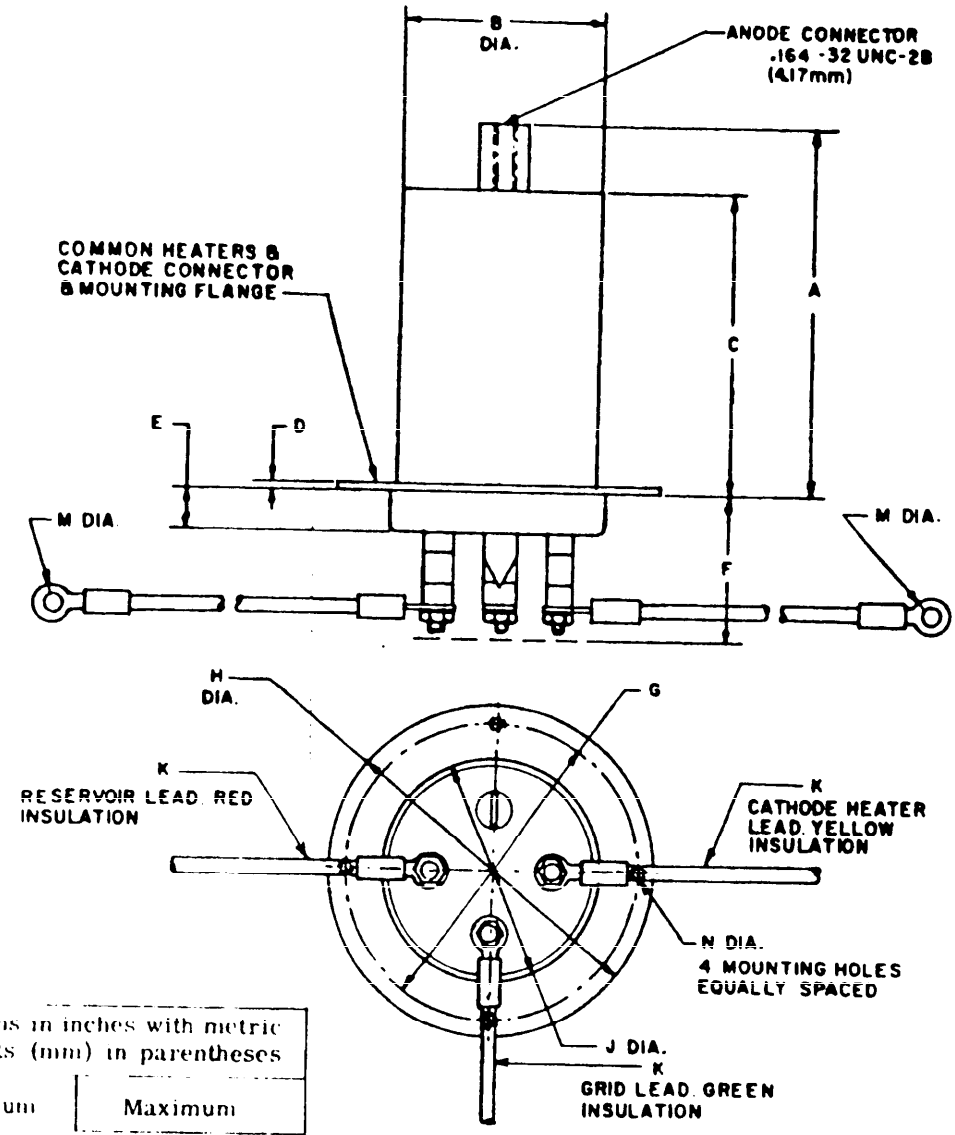
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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses.	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	2.43 (61.72)	2.47 (62.74)
(D) B	1.740 (44.20)	1.760 (44.70)
C		.28 (7.11)
E	1.25 (31.75)	
(D) F	2.245 (57.02)	2.255 (57.28)
G		1.50 (38.10)
Reference dimensions		
D	.030 (.76)	
H	2.031 (51.59)	
J	6.00 (152.40)	
K	.203 (5.16)	
M	.147 (3.73)	

FIGURE 1. Outline drawing of electron tube type 7665.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	2.377 (60.38)	2.497 (63.42)
B	1.365 (34.67)	1.385 (35.18)
C	1.877 (47.68)	1.997 (50.22)
E		.279 (7.04)
F		1.00 (25.40)
H		2.250 (57.15)
J		1.50 (38.10)
Reference dimensions		
D	.030 (.76)	
G	2.031 (51.59)	
K	6.00 (152.40)	
M	.203 (5.16)	
N	.147 (3.73)	

FIGURE 1A. Outline drawing of electron tube type 7665A.

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